

TITLE PAGE

- **Problem Statement ID** – SIH26181
- **Problem Statement Title**- AI-Powered Personal Health Companion & Edge Disaster Monitor
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Hardware
- **Team ID**- 171695
- **Team Name**- CHIPSTERS



VALOR: VITAL AND ATMOSPHERIC LOGIC FOR OFFLINE RESCUE

❖ AI-Powered FPGA accelerated Personal Health Companion & Edge Disaster Monitor

✗ How Current Technology Fails During Disasters

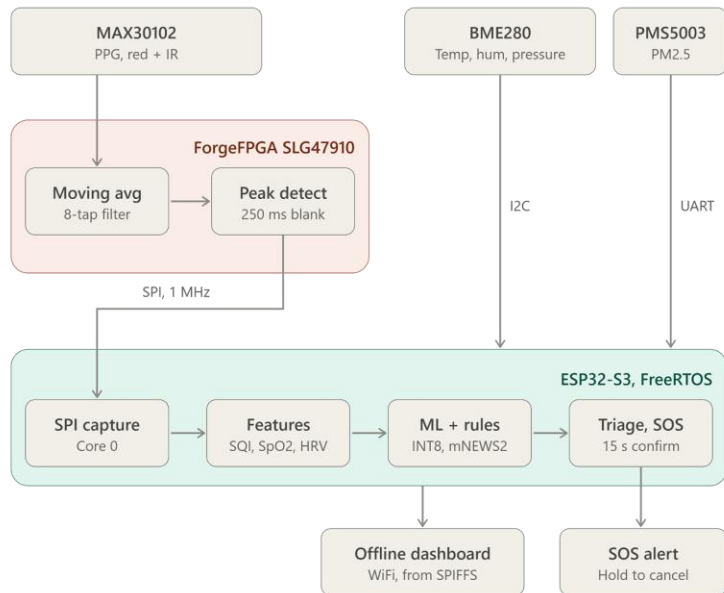
- **Wearables Are Environmentally Blind:** Commercial smartwatches track fitness steps and resting heart rate, but have zero awareness if ambient air is toxic (AQI 400~500+) or if 48°C heat is causing heat exhaustion so it can't connect the dots between these conditions, especially when outages shut down network towers
- **Environmental Sensors Ignore the Body:** Weather apps report regional city forecasts, not what an individual worker's cardiovascular system is actually enduring at street level.
- **The Cloud Dependency Trap:** Cloud AI health platforms require mobile networks. During floods, cyclones, or power grid failures, cellular towers collapse and cloud health apps die completely.
- **Software Jitter Destroys HRV:** Operating system scheduling delays on standard CPUs jitter by 5-20 ms, severely corrupting heart rate variability (HRV) metrics.

✓ VALOR Proposed Solution & Innovation

- **Detailed Solution Explanation:** Simultaneously acquires dual-wavelength optical PPG pulses along with hyper-local ambient temperature, relative humidity, and laser PM2.5 particulate matter on the wrist.
- **How It Addresses the Problem:** Fuses bodily vitals with environmental stress in real time to calculate cardiac drift and respiratory strain, giving early warning of heat stress and abnormal vital signs before they become emergencies.
- **Hardware Innovation (FPGA Silicon):** Custom Verilog 8-tap filter and beat detector run in FPGA hardware at 50 MHz, taking time-critical filtering off the OS and removing scheduler jitter, with no DSP multipliers or BRAM.
- **Uniqueness & Zero Cloud Dependency:** Quantized INT8 neural network fits into only 619 bytes of INT8 weights, so privacy-preserving on-device inference works offline during blackouts.

Architecture, Stack & RTL Verification

HARDWARE & FIRMWARE FLOW



Survival UI: HTML5 • Vanilla JS • Canvas API • Win32 GDI

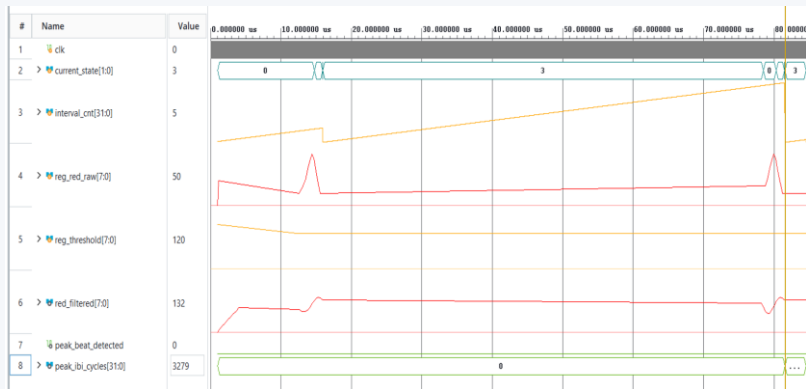
Clinical AI / TinyML: PyTorch • QAT INT8 • Scikit-Learn • C99

Firmware / OS: C99 • FreeRTOS • ESP-IDF v5.5.5

Hardware RTL: Verilog-2001 • SystemVerilog • 4-wire SPI

EDA / compilers: Vivado • ForgeFPGA Workshop • Icarus • GCC

RTL SIMULATION: reg_red_raw → 8-TAP FILTER → reg_filtered



Testbench, compressed time scale. Pulses ~66 μ s apart; peak_ibi_cycles = $3279 \times 20 \text{ ns} = 65.6 \mu$ s.

FEASIBILITY ANALYSIS: SILICON & ECONOMICS

Vendor-Agnostic Verilog RTL

Identical synthesizable core Verilog runs on AMD Xilinx (6-LUT) and Renesas ForgeFPGA (5-LUT) architectures without redesign.

Xilinx + ForgeFPGA

Ultra-Low Silicon Footprint

Synthesized on Renesas ForgeFPGA requiring only 364 / 1120 LUTs with 0 Block RAM.

32.5% Utilization

Future Sensors

IMU for falls, activity and sleep; skin-temperature sensor; GPS module for SOS location.

Planned

Economic Viability (< \$15 / ~₹1,250 Bulk BOM)

12-component commodity bill of materials (ForgeFPGA + ESP32-S3 + 3 sensors + OLED). Zero recurring subscription.

Zero Cloud Fees

RISKS & MITIGATION STRATEGIES

Motion & Optical Noise

Mitigation: 8-tap running-sum filter on FPGA + Karlen SQL algorithm to discard motion-corrupted pulses.

Micro-FPGA Logic Limits

Mitigation: Pure-logic O(1) bit-shift division ($>> 3$), requiring 0 DSP multipliers and 0 BRAM on ultra-low-cost FPGAs.

Grid & Cellular Blackouts

Mitigation: 100% on-device offline TinyML (619 bytes INT8 weights) and local SoftAP web status server.

Clinical Accuracy

Mitigation: Vital-sign triage evaluated offline on 16,387 MIMIC-III ICU records (94.11% accuracy). ICU-grade data; wrist-worn field validation is the next step.

Future Features

Roadmap: Fall detection, activity & sleep tracking, caregiver SOS with GPS location; Snapdragon Wear W5+ Gen 1 port.

50 MHz

cycle-accurate FPGA
pipeline (20 ns clock)

94.11%

vital-sign triage accuracy,
16,387 MIMIC-III records

619 B

offline INT8 network
(6 → 24 → 16 → 3)

12

components in the
entire wearable BOM

SOCIAL IMPACT

• Offline Triage

Operates reliably during severe cellular
network and power grid blackouts.

• Complete Privacy

Sensitive health vitals never leave device;
zero cloud dependence & no smartphone
required.

• Early Warning & SOS

Heat-stress, air-quality and vitals alerts for
vulnerable groups, plus SOS.

ECONOMIC VIABILITY

• Low-Cost Hardware

Built using 12 off-the-shelf components.
Bulk BOM < \$15 (~₹1,250) | Fully assembled
< ₹2,500.

• Zero Overhead

No cloud server fees, no mobile data plans,
and zero recurring subscription costs.

• Scalable Deployment

One edge-AI pipeline for individuals,
providers and disaster agencies.

ENVIRONMENTAL SUSTAINABILITY

• Zero-Cloud Footprint

Triage runs on the wrist, so it needs no
data-center processing or network
bandwidth.

• Pure Logic FPGA

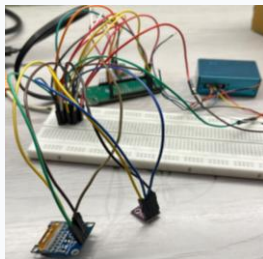
Uses 0 DSP blocks and 0 BRAM, keeping
the FPGA footprint and power draw low.

• Disaster-Specific Alerts

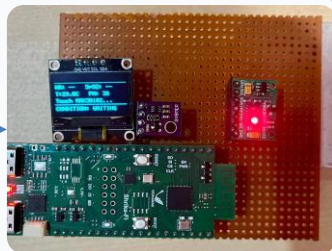
Heat, pollution and flood risk scores, plus
cyclone warning from pressure trend.

IN Tailored for Indian Disasters: **Protects 380M+ outdoor workers and elderly and rural citizens** across North India heatwaves (48°C)
• Toxic smog (AQI 400–500+) • Monsoon flash floods & blackouts

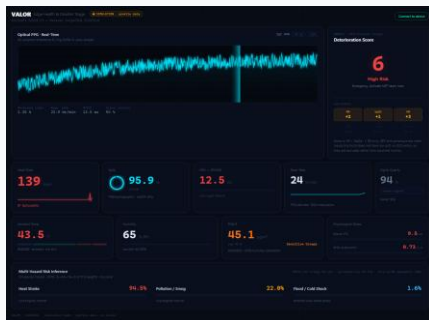
BULKY PROTOTYPE TO COMPACT ZERO PCB DESIGN



BULKY BREADBOARD PROTOTYPE



COMPACT ZERO PCB



Live 60 FPS Disaster Dashboard

- 100% Offline SPIFFS Web Server
- Real-time optical systolic trace
- RSA breathing wave extraction
- Live mNEWS2 & multi-hazard radar

PEER-REVIEWED CLINICAL & ENGINEERING REFERENCES

CLINICAL / TECHNICAL STANDARD (PEER-REVIEWED)	IMPLEMENTED IN WHICH OF OUR CODE FILES
Royal College of Physicians (2017) NEWS2 Acute Severity	clinical_vitals_engine.c
Charlton PH et al. (IEEE 2018) EDR Breathing Rate from PPG	ppg_respiratory_rate.c
Karlen W et al. (Physiol Meas 2012) Signal Quality Index (SQI)	ppg_sqi.c
Moran DS et al. (Am J Physiol 1998) Physiological Strain (PSI)	disaster_risk_engine.c
Brook RD et al. / AHA (Circulation 2010) PM2.5 Cardiac Stress	disaster_risk_engine.c
Johnson AEW et al. (MIMIC-III 2016) 16,387 ICU Records	data/mimic/ (Host Evaluator)
Si M et al. (Atmos Meas Tech 2019) ML Particulate Calibrator	pm25_calibration_int8.c
Steadman RG (J Appl Meteorol 1979) Sultriness & Heat Index	disaster_risk_engine.c
SPI Mode 0 Protocol Specification	forgepga_ppg_top.v